MindCube: Spatial Mental Modeling from Limited Views

Group Number: 11

Weekly Report 2

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Problem Statement: Guiding Vision–Language Models to Generate Cognitive Maps and Reason Over Them.

1 Introduction

The project is about the spatial question answering benchmark designed to evaluate spatial mental modeling from limited multi-view observations. This project aims to systematically create VLMs built on robust spatial mental models by representing positions (cognitive mapping), orientations (perspective-taking), and dynamics (mental simulation of "what-if" movements).

2 Work Done (Week 2):

During the first week, our group focused on the following tasks:

- Understanding the baseline model architecture
- Understanding the Dataset

2.1 Architecture

The baseline model provided for this problem statement is a VLM (Vision-Language Models) which can form spatial cognitive maps from multi-view images. The model connects unseen spatial relationships to answer the spatial reasoning questions. First, it builds a cognitive map based on space.

There are 3 major modules in the Vision Language models:

Vision Encoder: This is utilized to extract visual information from the image such as features which represent objects, their orientations etc. Multi-View Fusion: Since the input dataset consists of multi-view images, the model needs to combine the information across the different views, which also helps to resolve the position/neighbourhood of the object. Cognitive Map: The model then generates a cognitive map which provides a graphical representation of the objects and relations. This graphical representation connects objects(nodes) to each other through edges(which depict spatial relation). Reasoning: Then the model performs reasoning over the graph according to the question to give an output.

This model jointly trains cognitive map generation and reasoning to obtain improved performance.

2.2 Dataset

MindCube features multi-view image groups paired with spatial reasoning questions, enabling fine-grained analysis of spatial modeling performance. It targets key challenges such as maintaining object consistency across views and reasoning about occluded or invisible elements.

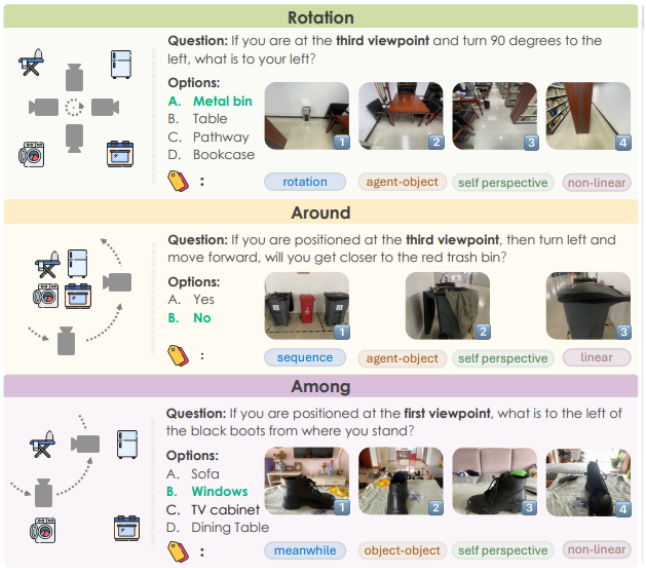

Figure 1: Camera Movement Patterns(Rotation, Around, Among) with corresponding spatial QA

The multi-view image dataset is taken using three types of camera movements: Rotation, Around and Among. In the Rotation type, the camera stays in place but rotates to look around the environment. In the Around type, the camera moves in a circular path, keeping the evaluated object in the centre.

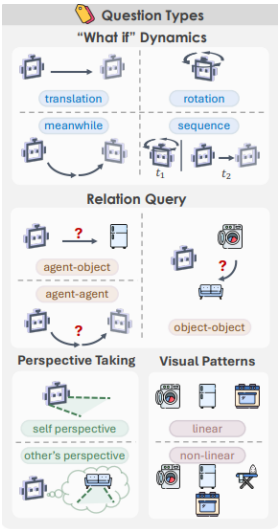

Figure 2: Spatial Question Types

This focuses on capturing different sides of the object. In the Among type, the camera moves along a circular path among the evaluated objects, focusing on each object against different backgrounds.

In the question types, the "What if" dynamics focuses on the hypothetical transformations applied to the agent's viewpoint, such as translation, rotation, or their combination (meanwhile and sequence). Relation Query focuses on the type of spatial relation being queried, including agent–object, agent–agent, or object–object. Perspective-taking is whether the spatial reasoning is grounded in the perceiver's own viewpoint (self) or involves adopting the viewpoint of another entity (other). While Visual Patterns describes the objects' spatial configurations, including spatial linear or non-linear arrangements.

3 Next Set of Work:

In the upcoming week, we plan to:

- Complete the model setup
- Deeply understanding the model architecture
- Understanding related work done from the literature.

4 Conclusion

During the second week, we tried to understand the model architecture and the dataset. While we got a brief idea of the baseline model, we're yet to fully

understand it and execute it.

References

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